# PEP-II

# RF SYSTEM

# RF STATION SAFETY SURVEY

# PS-340-330-55-R3

Submitted by _____ Date 1/4/06 Klystron Department
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Approved by _____ Date 10-2-06 Accelerator Department
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Approved by _____ Date 1-5-06 Accelerator Department
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RF STATION SAFETY SURVEY

Stanford Linear Accelerator Center
Revised: December 2, 2005

DCR

Station:_____ Date:_____ Initial:_____

This procedure is to be used when a klystron is installed in an IR. Records shall be kept by the Klystron RF Engineer.

Survey performed by the PEP-II RF Group or SLAC AMRF Maintenance Group. RP Field Operations Group (X4299) needs to be contacted to perform ionizing radiation survey.

A safety survey needs to be performed for the RF station once a year or after major downtimes of several days duration.

| **Item** | **Verified (Y/N)** | **Initials** | **Date** |
|---|---|---|---|
| Waveguide connections made up, check torque on 6 random bolts > 25 ft-lbs on all accessible flanges in the surface building and in the ring tunnel (see PS-340-330-61-R3 PEP-II RF Non-Ionizing Radiation Safety Procedure, Section 6) | | | |
| Waveguide Pressure Interlock Test: Verify interlock operation (see PS-340-330-61-R3 PEP-II RF Non-Ionizing Radiation Safety Procedure, Section 11) | | | |
| Klystron grounding cable connected | | | |
| Klystron HV cable connected with 4 screws in place | | | |
| Verify that the HVPS Hoffman Box cover is in place and securing bolts are utilized | | | |
| Klystron lead shielding and collector lead panels in place per klystron traveler photos and documentation | | | |
| Klystron area posted as Radiation Area (RA) | | | |

| | | | |
|---|---|---|---|
| **<u>Before</u> RF station turn ON and while this station survey is being completed, ALL personnel occupying within and near the posted RA around the klystron must be RWT-I certified.** | | | |

**<u>Radiation Surveys at 100 kW</u>**

With the RF station turned on at a klystron output power level of <u>approximately 100 kW</u>, a RF non-ionizing radiation survey using a calibrated RF field meter is performed wherever access dictates. The RF radiation level must be below 0.1 mW/cm$^2$ at every accessible waveguide joint. In addition, the RP should perform an ionizing radiation survey around the klystron to check if sufficient lead shielding is installed to keep radiation below 5 m<u>R</u>/hr at 30 cm away from any surface of the klystron and < 100 mR/h on contact.

| **Item** | **Klystron Power Level (kW)** | **Max Radiation Level (mW/cm$^2$)** | **Initials** | **Date** |
|---|---|---|---|---|
| Waveguide Non-Ionizing Radiation | | | | |
| **Item** | **Klystron Power Level (kW)** | **Max Radiation Levels at 30-cm and contact (mR/hr)** | **Initials** | **Date** |
| Klystron Ionizing Radiation | | | | |

**Radiation Survey at Full Power**

The RP Field Operations Group should perform an ionizing radiation survey around the klystron at its final operating level to check if sufficient lead shielding is installed to keep radiation below 5 mR/hr at 30 cm away from any surface of the klystron and < 100 mR/h on contact.

| Item | Klystron Power Level (kW) | Max Radiation Levels at 30-cm and contact (mR/hr) | Initials | Date |
|---|---|---|---|---|
| Klystron Ionizing Radiation | | | | |

**Revisions**

| Date | Revision | Comments |
|---|---|---|
| 6/08/99 | R1 | Unknown changes |
| 1/20/05 | R2 | Format changed to include Date for each checklist item.<br><br>Accelerator Department Deputy Head for PEP-II Beams and Safety Officer included on cover sheet.<br><br>Clarified that waveguide pressurization interlocks are to be checked.<br><br>Included HVPS Hoffman box cover verification.<br><br>Added that RWT I classified personnel ONLY allowed in PEP-II RF building while station certification is being completed.<br><br>Included klystron ionizing radiation survey at 100 kW. |
| 12/2/2005 | R3 | Added the RP Field Operations Group is to do the Ionizing Radiation survey at 100-kW and at the final operating level.<br><br>Added that the klystron lead shielding and collector lead panels are to be verified per the photos and documentation in the klystron traveler.<br><br>Added the posting of RA. |
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